

Microprocessor Systems

Lab7-1 IR Distance Measuring Sensor-GP2Y0A51SK0F
& Analog Digital Convertor-MCP3202

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NTHU EE



Introduction

- GP2Y0A51SK0F is a distance measuring sensor unit.
- Composed of an integrated combination of
 - PSD (position sensitive detector)
 - IR-LED (infrared emitting diode)
 - Signal processing circuit



Features

- Short measuring cycle (16.5ms)
- Distance measuring range : 2 to 15 cm
- Compact size (27.0 × 10.8 × 12.0mm)
- Analog output type

IR Sensors Key Applications

- TVs, computers, laptops
- Distance measurement (in this lab)
- Security systems such as surveillance, burglar alarms, etc.
- Monitoring and control applications

IR Sensors Key Applications

Surveillance
Security



Mobile
Biometric Systems



Digital
Healthcare



Gaming
Notebook



Large Size Touch
Panel



NIR
Spectroscopy



AR, VR
Eye Tracking



Drones



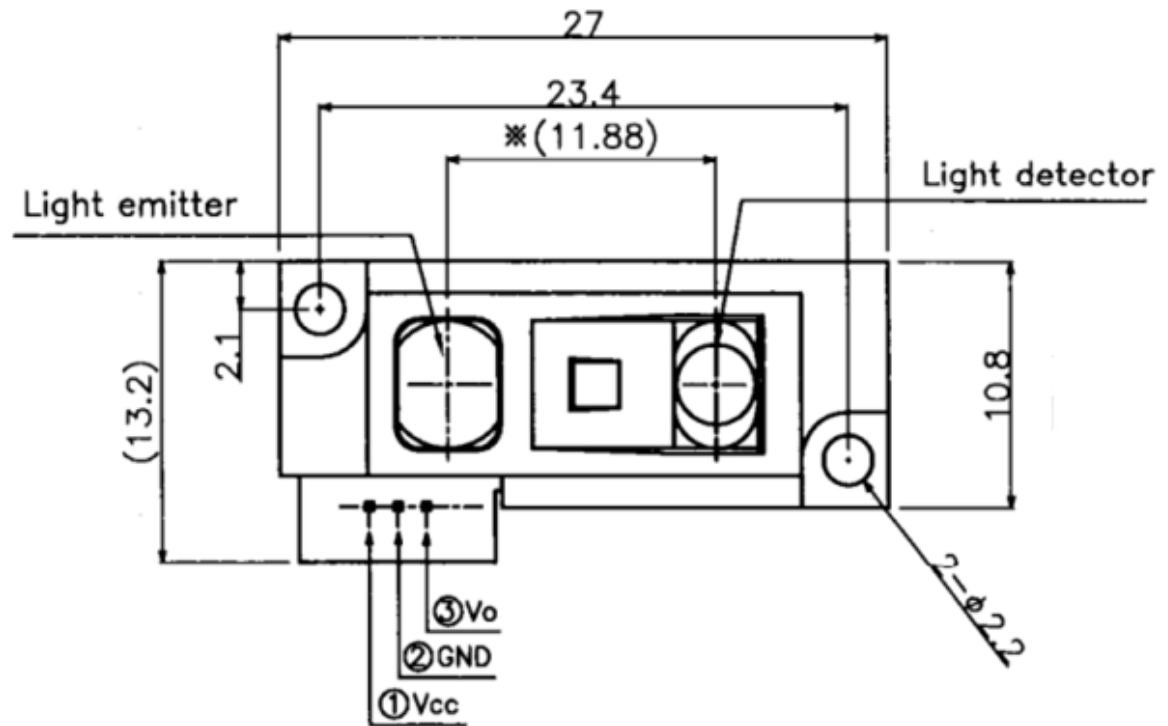
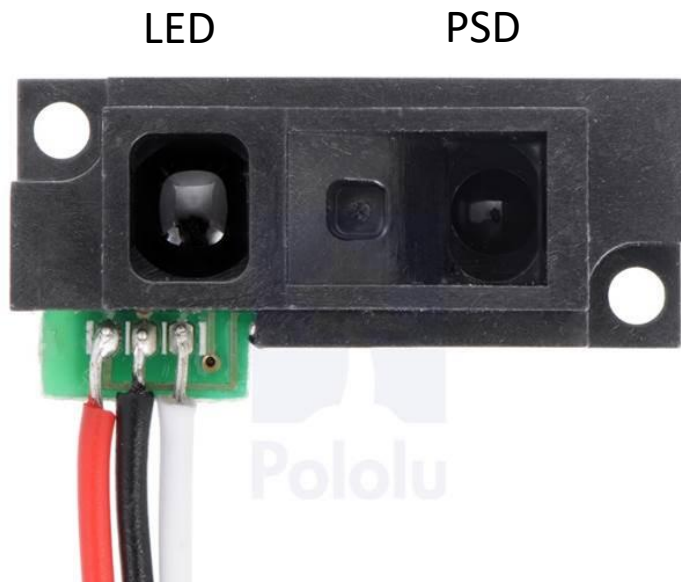
Car Sensors



LIDAR
Technology

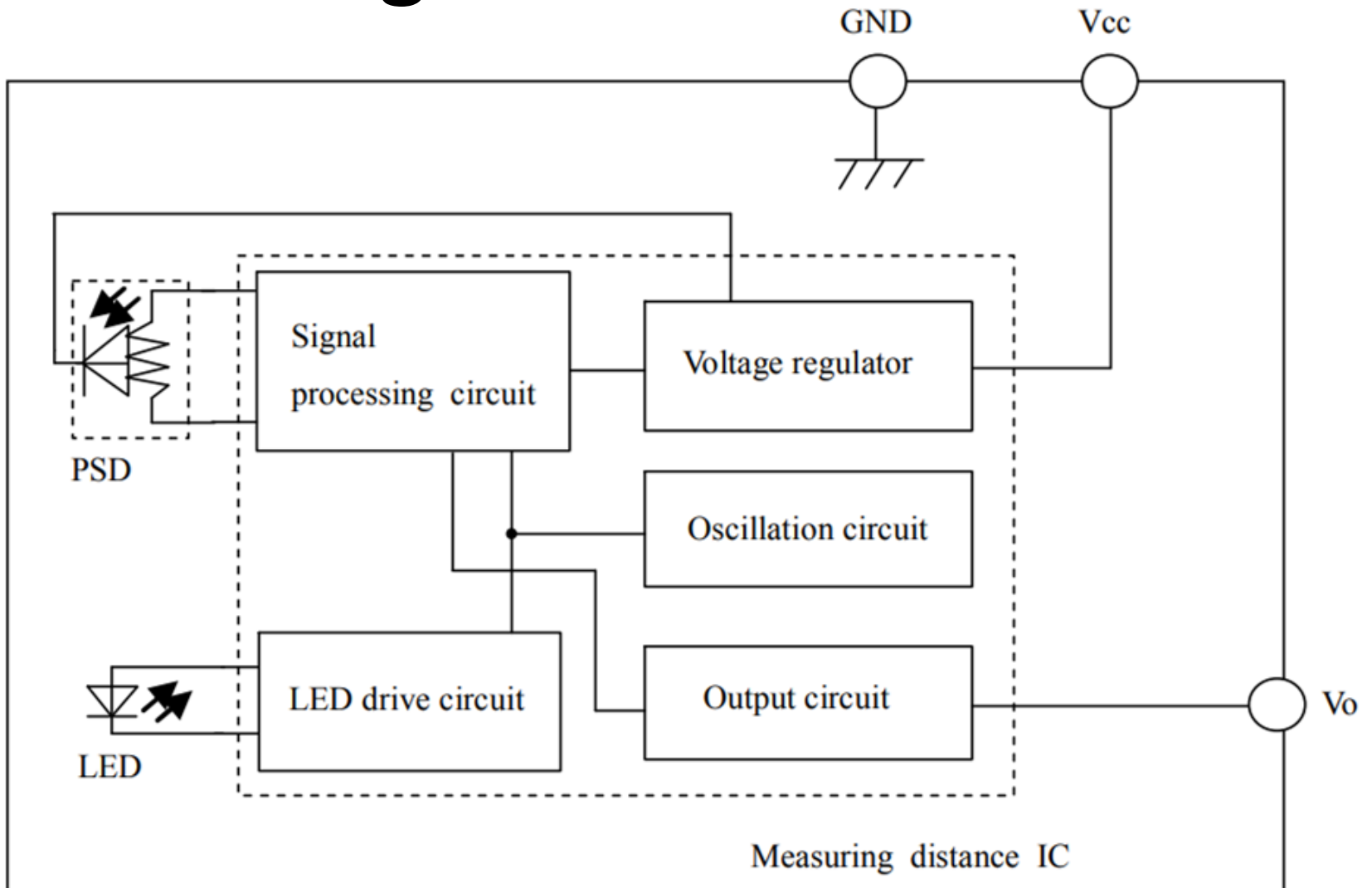


Appearance

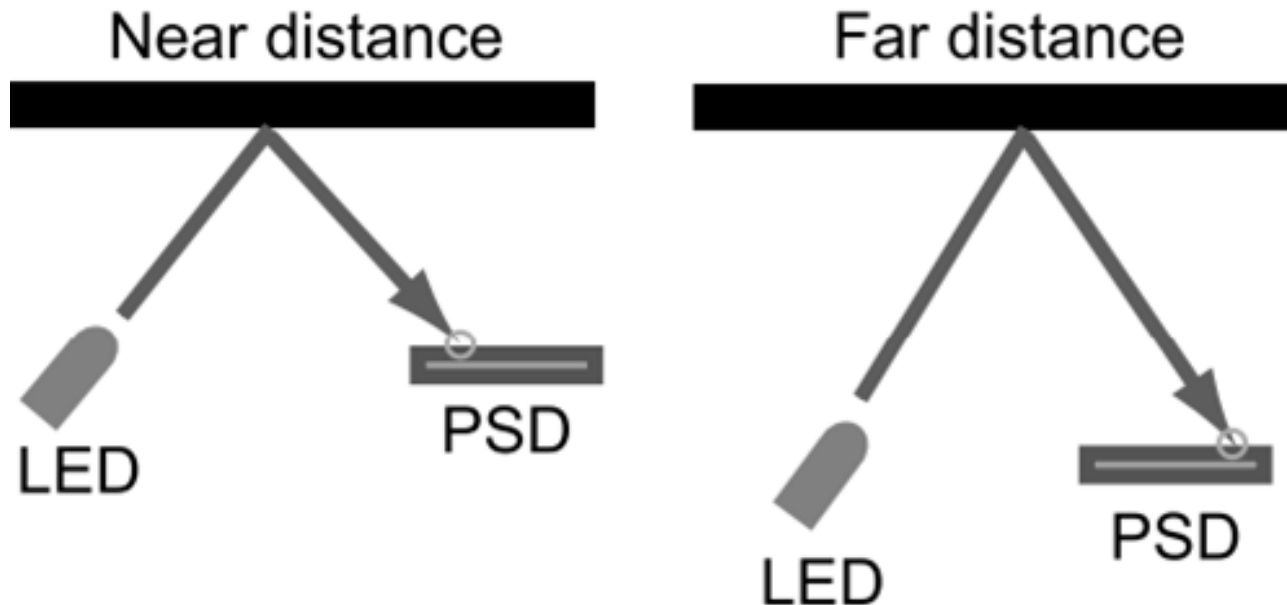


- An IR LED emitter lens that emits a light beam.
- A position-sensible photodetector (PSD) where the reflected beam will fall onto.

Block Diagram

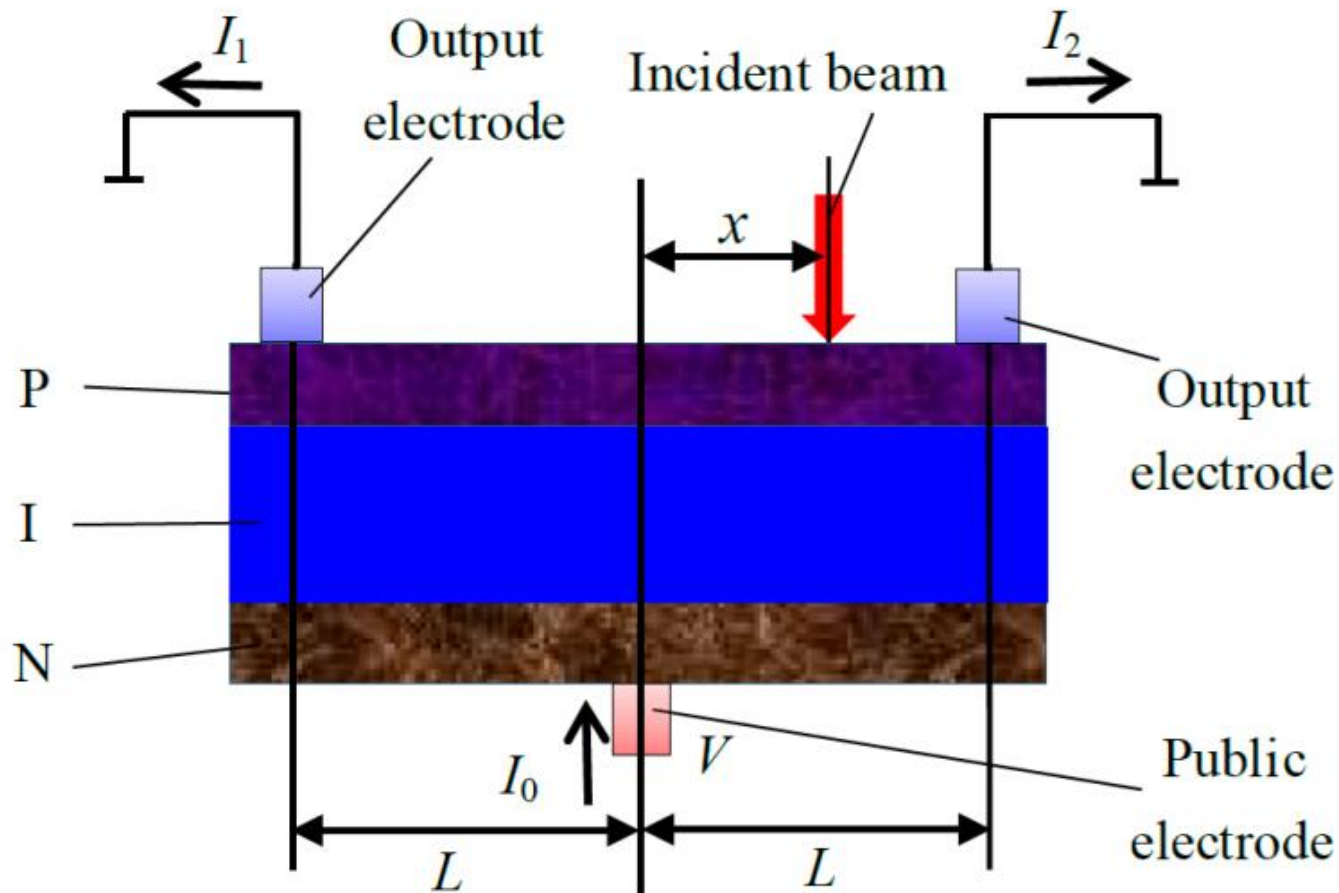


Basic Working Theory



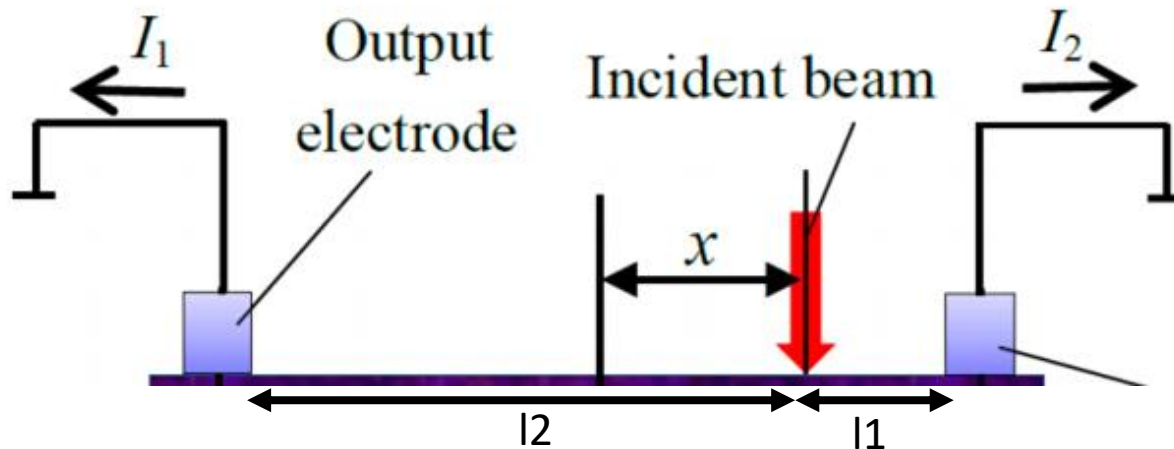
- This device outputs the voltage corresponding to the detection distance.

Position Sensitive Detector



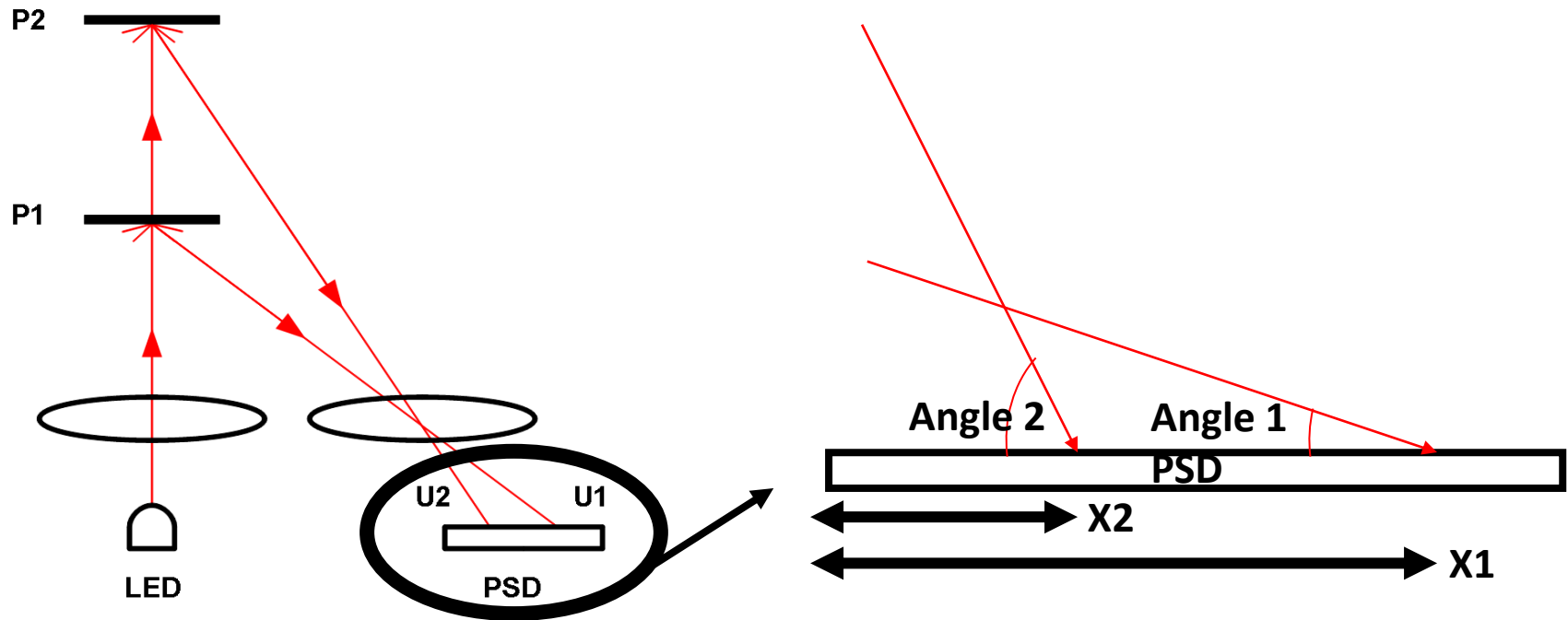
Position Sensitive Detector

- The two output electrodes receive different intensities, I_1 and I_2 , which are based on the incident beam position x .
- The incident beam position x can be calculated with the formula below, and further used to determine the incident light spot distance.



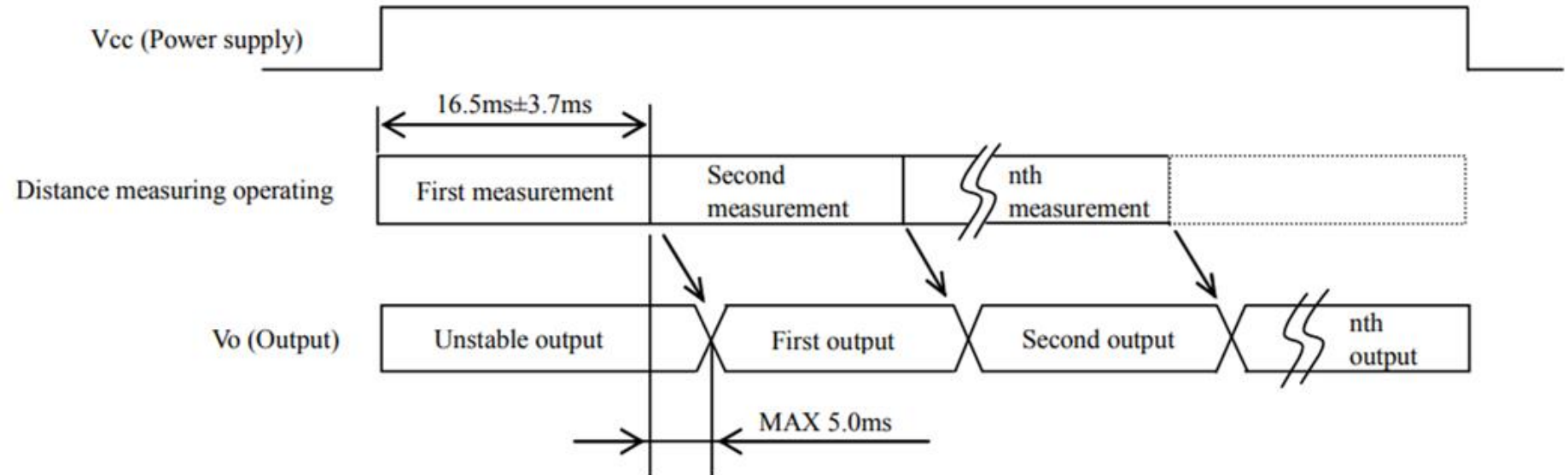
$$x = L * (I_2 - I_1) / (I_2 + I_1)$$

Triangulation Measurement



- The angle between reflective light and PSD can be derived knowing the centroid position of reflective light(X).
- The distance between object and device can then be calculated by triangulation.

Timing Chart



- Measuring cycle 16.5ms
- Max output delay 5ms

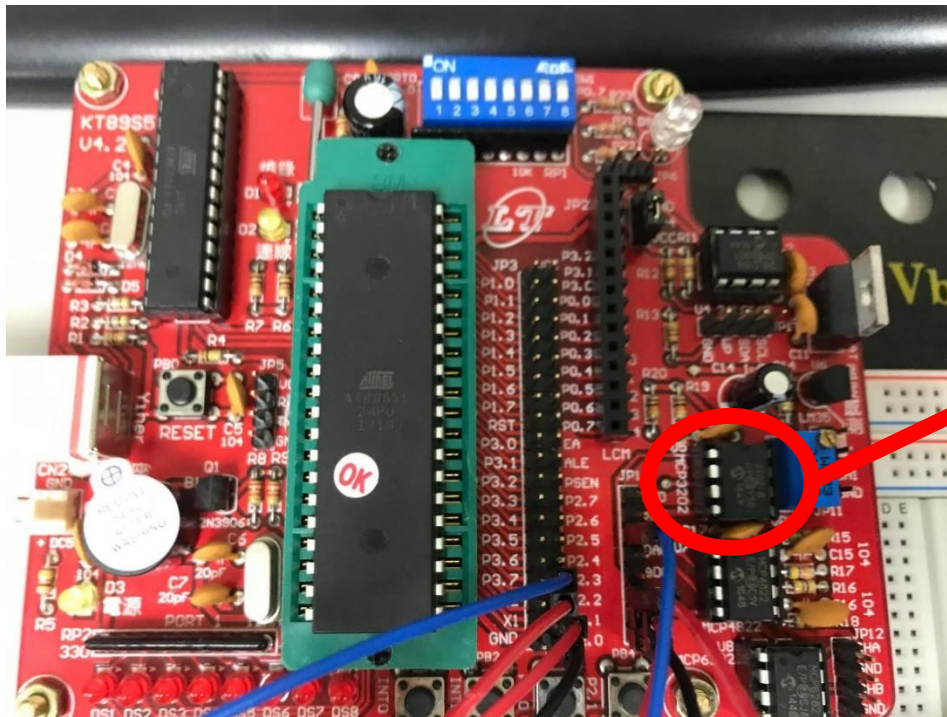
Example of Output



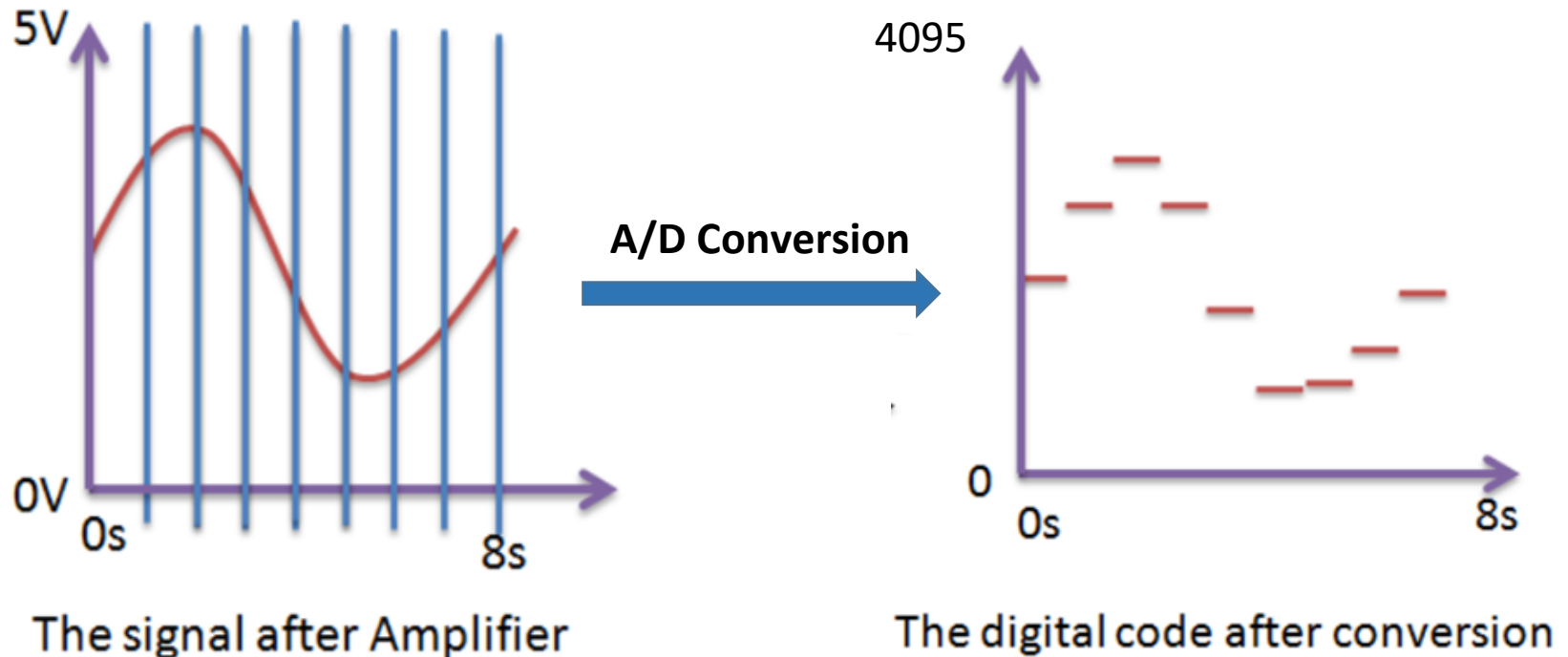
- X-axis: Distance to the reflective object(cm).
- Y-axis: Analog voltage output(V)
- Output voltage drops with greater distance.

Analog Digital Converter-MCP3202

- We need an ADC to convert the analog output from the IR distance sensor to digital values.
- MCP3202 is a 12-bit resolution ADC



ADC Introduction



- Sampling rate for MCP3202
 - Maximum 100 ksps at $V_{DD} = 5V$
 - Maximum 50 ksps at $V_{DD} = 2.7V$

N-bit Output Code

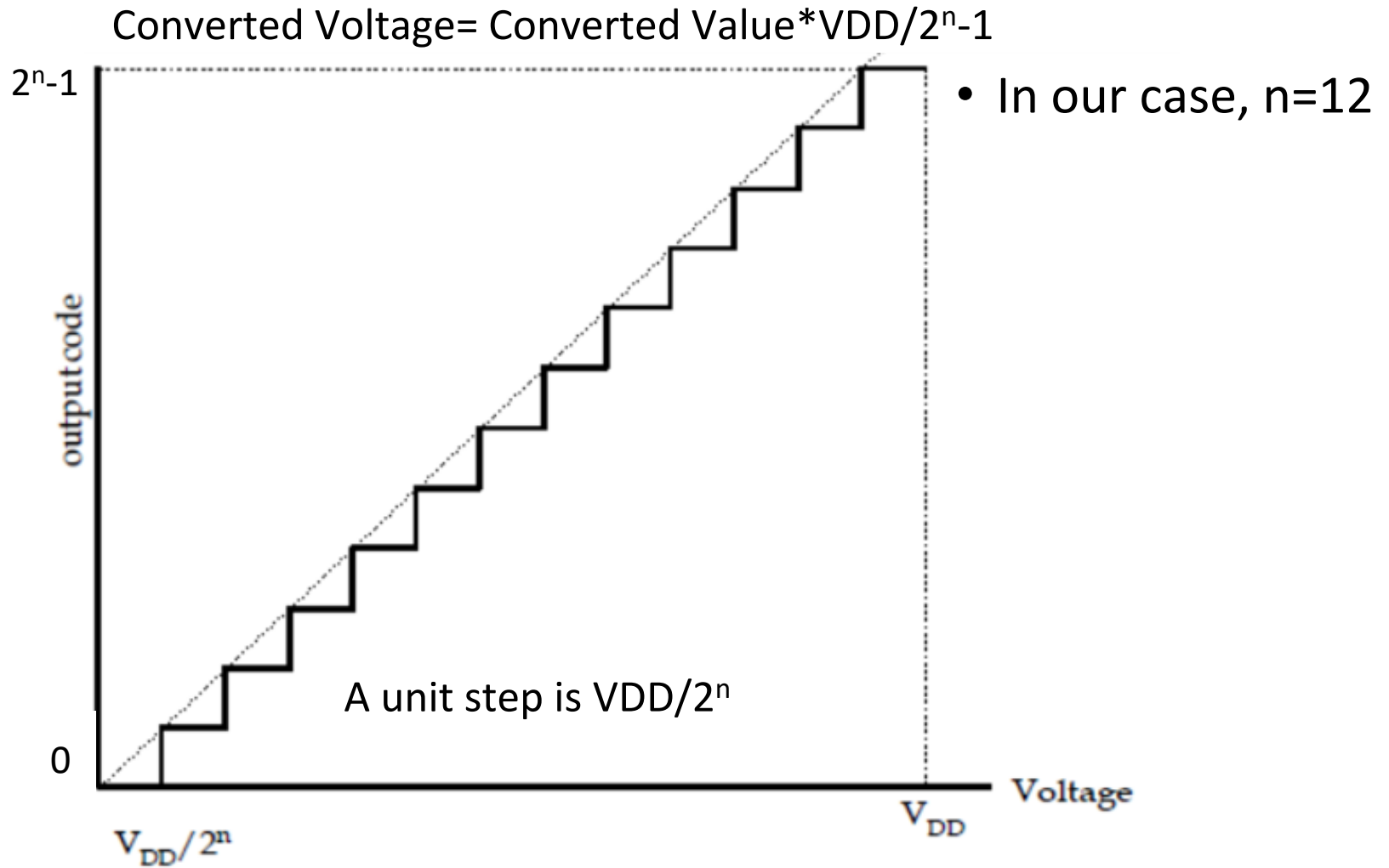
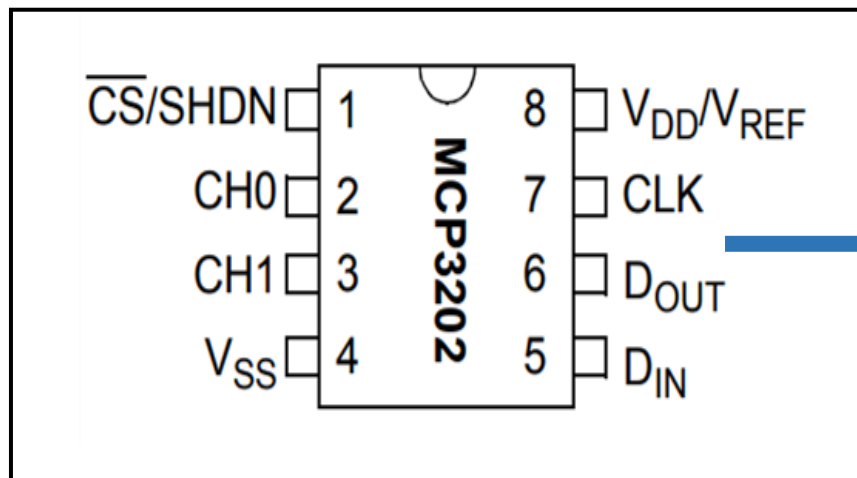


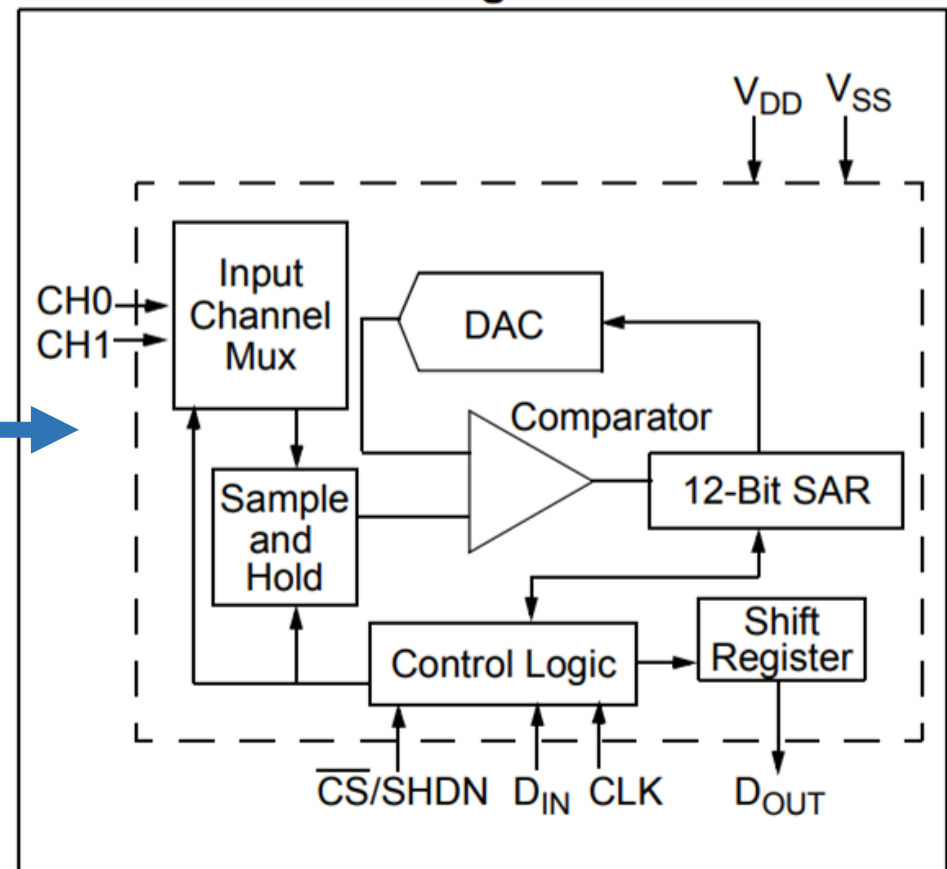
Figure 12.3 Output characteristic of an ideal n-bit A/D converter

Block Diagram

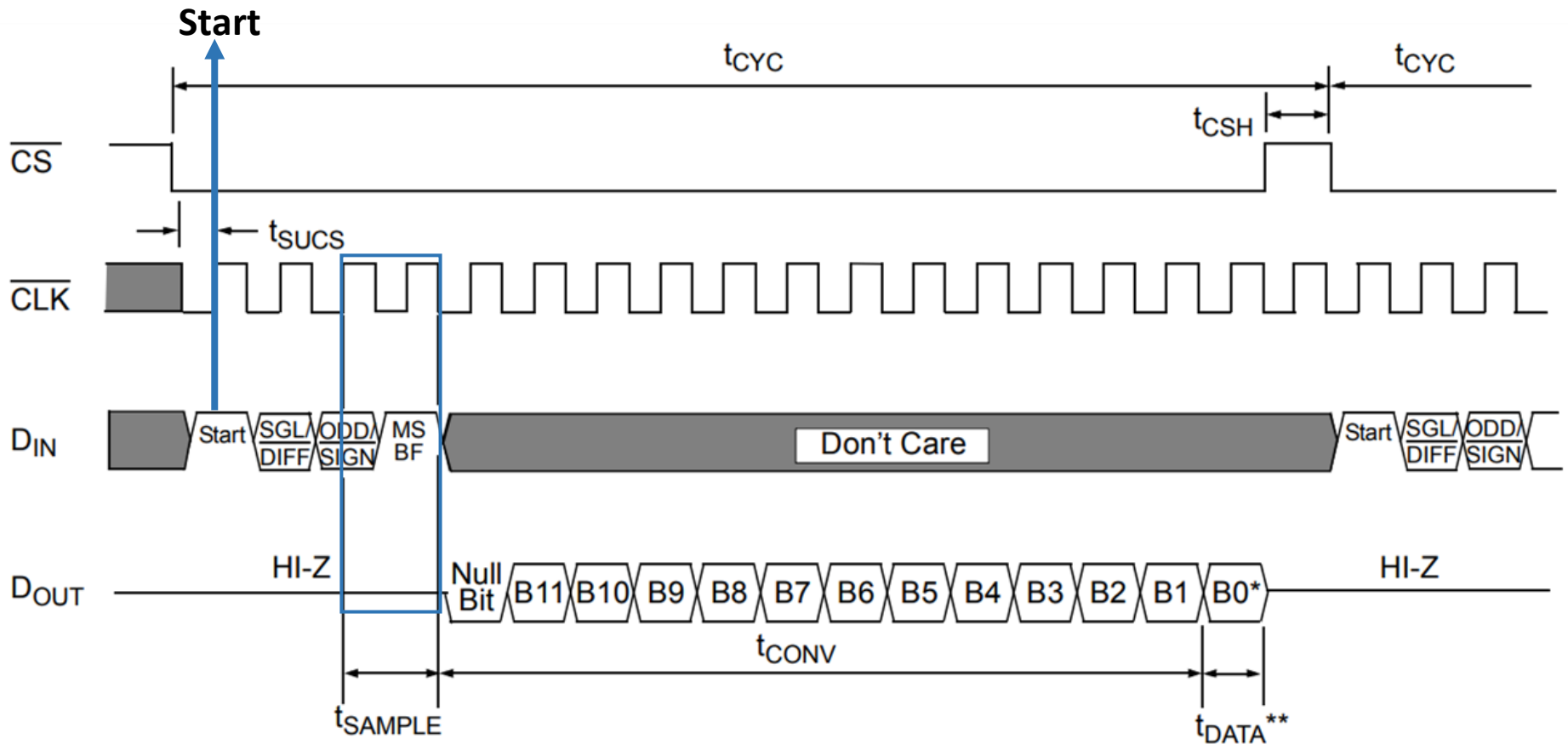
- Using SPI interface, same as MAX7219 in LAB2



Functional Block Diagram

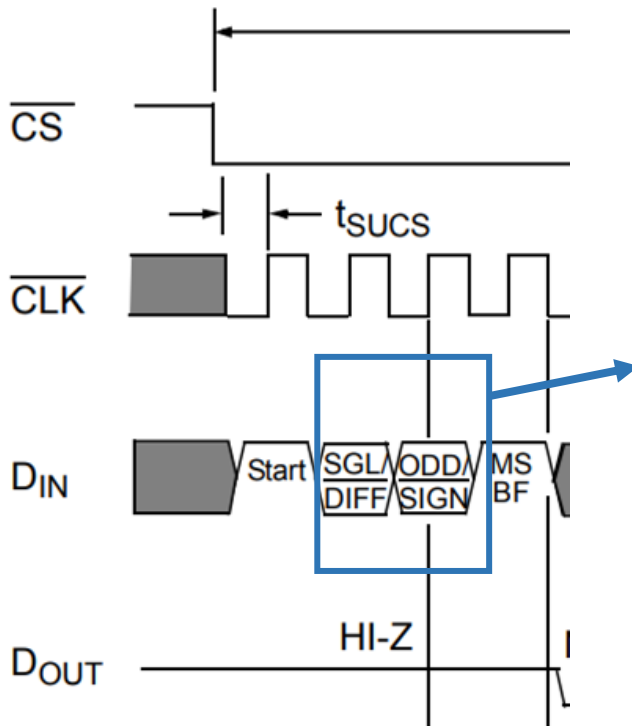


Timing Diagram



- The first D_{IN} with $\overline{CS}$ low consists of a start bit.
- Sampling time = 1.5 cycles.

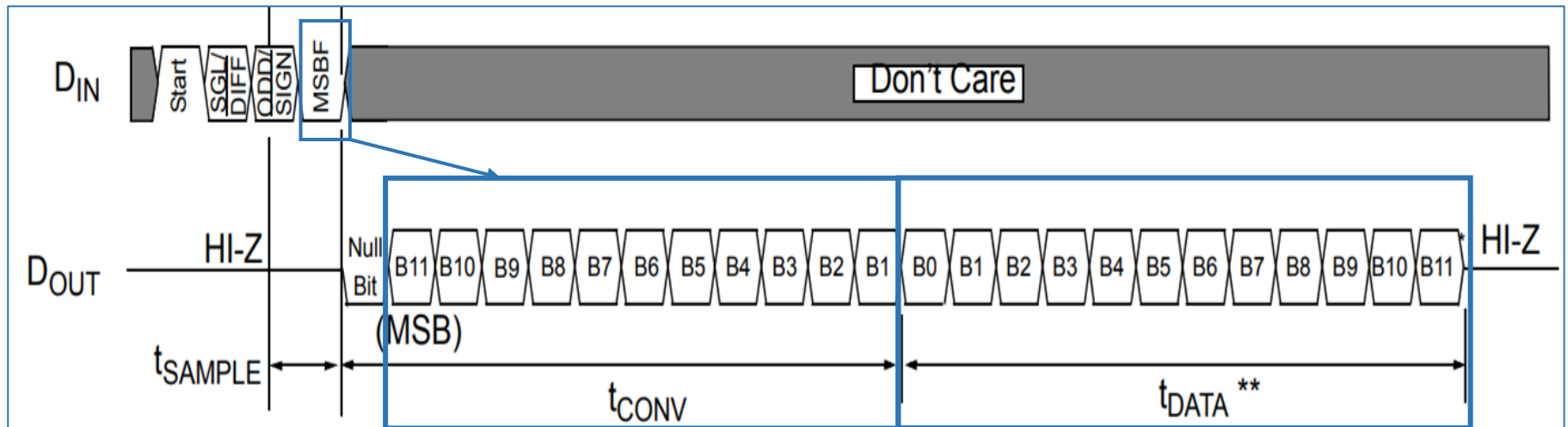
Work Mode



	Config Bits		Channel Selection		GND
	<u>SGL/</u> <u>DIFF</u>	<u>ODD/</u> <u>SIGN</u>	0	1	
Single-Ended Mode	1	0	+	—	-
	1	1	—	+	-
Pseudo-Differential Mode	0	0	IN+	IN-	
	0	1	IN-	IN+	

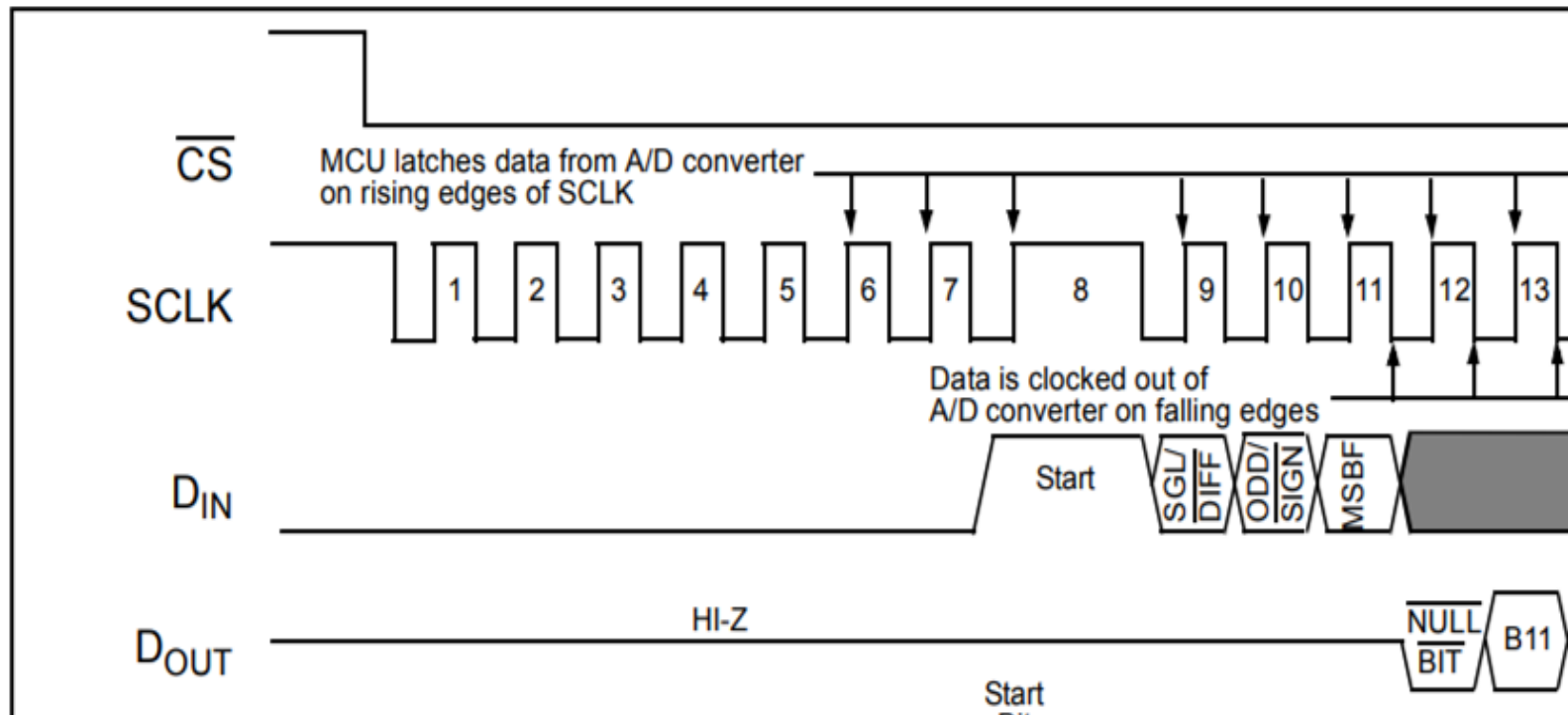
- Single-ended(1)/Channel select(0 or 1)
- Pseudo-Differtial(0)/Polarity(+ or -)

Work Mode



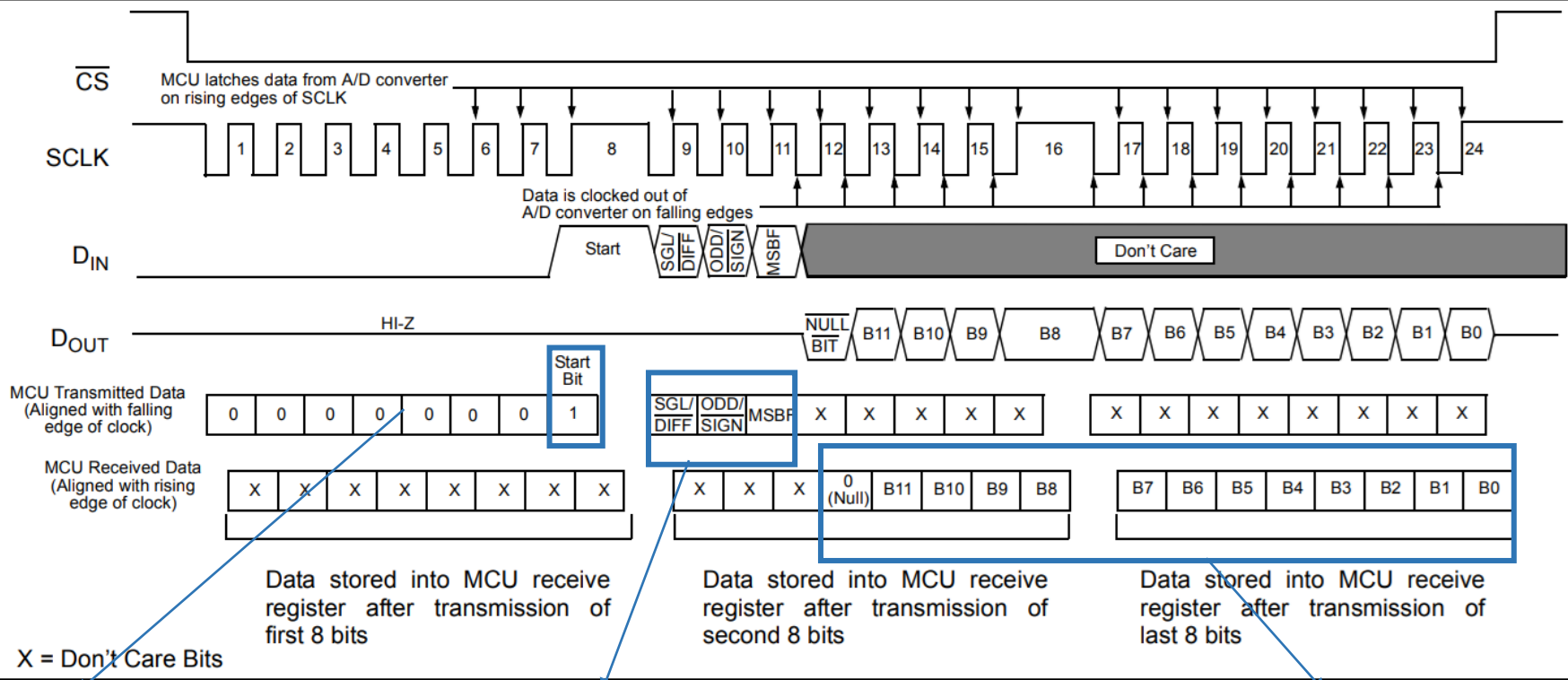
- MSBF bit is transmitted and used to enable the LSB-first format.
- If the MSBF bit is low, the device will output the converted word LSB first after the word has been transmitted in the MSB-first format.

Communicate with 8 bit MCU



- Data is transmitted from ADC on falling edges of SCLK.
- Data is received by MCU on rising edges of SCLK.

Communicate with 8 bit MCU



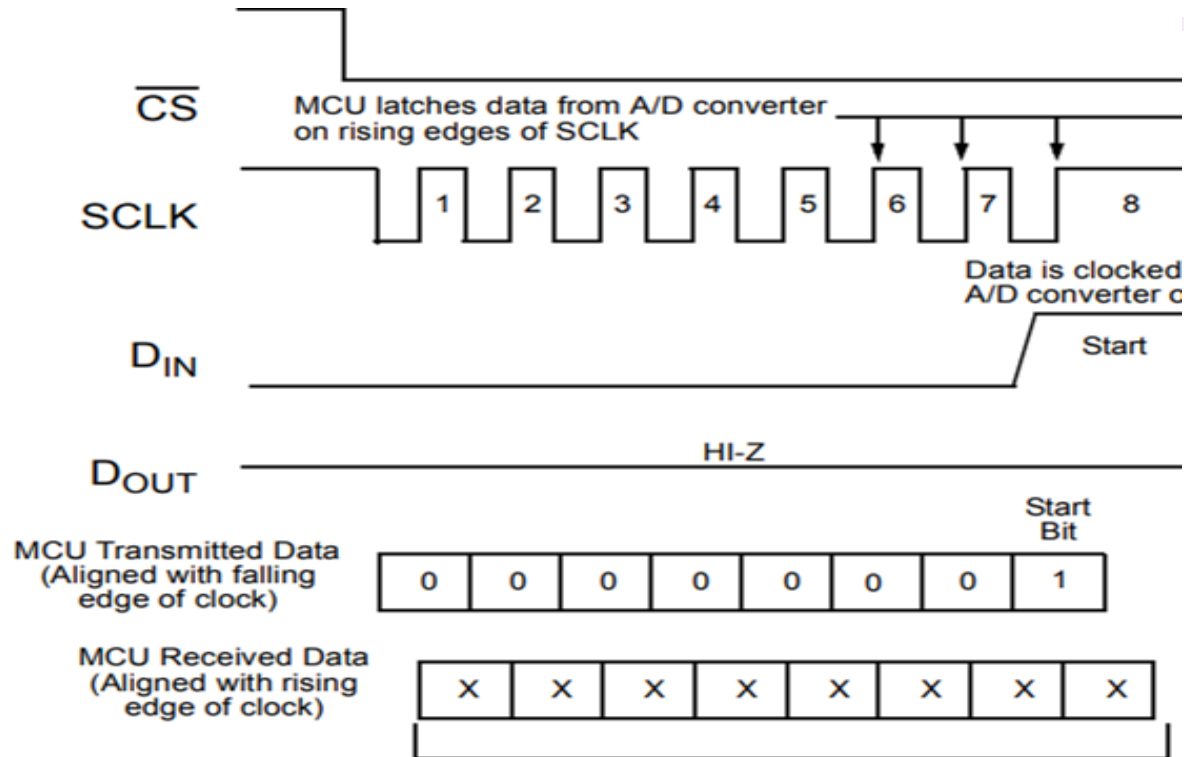
1bit start

2bit choose mode+1bit MSBF

1bit null+12bit ADC output

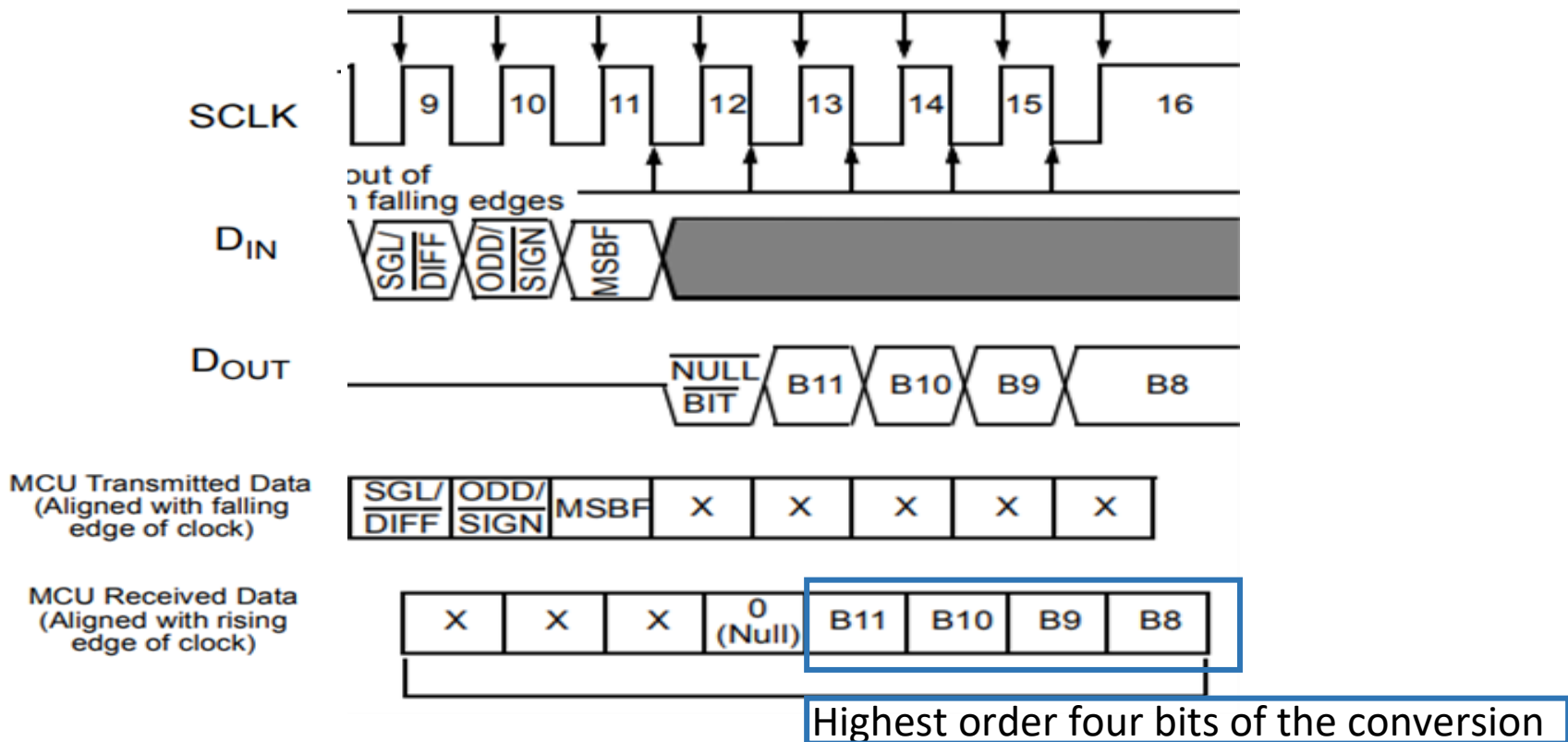
- It is required to send groups of 8 bits, how do we deal with it?
- In order to get these 17 bits, we need $8 \times 3 = 24$ cycles.

8 Bit MCU Transmission Arrangement



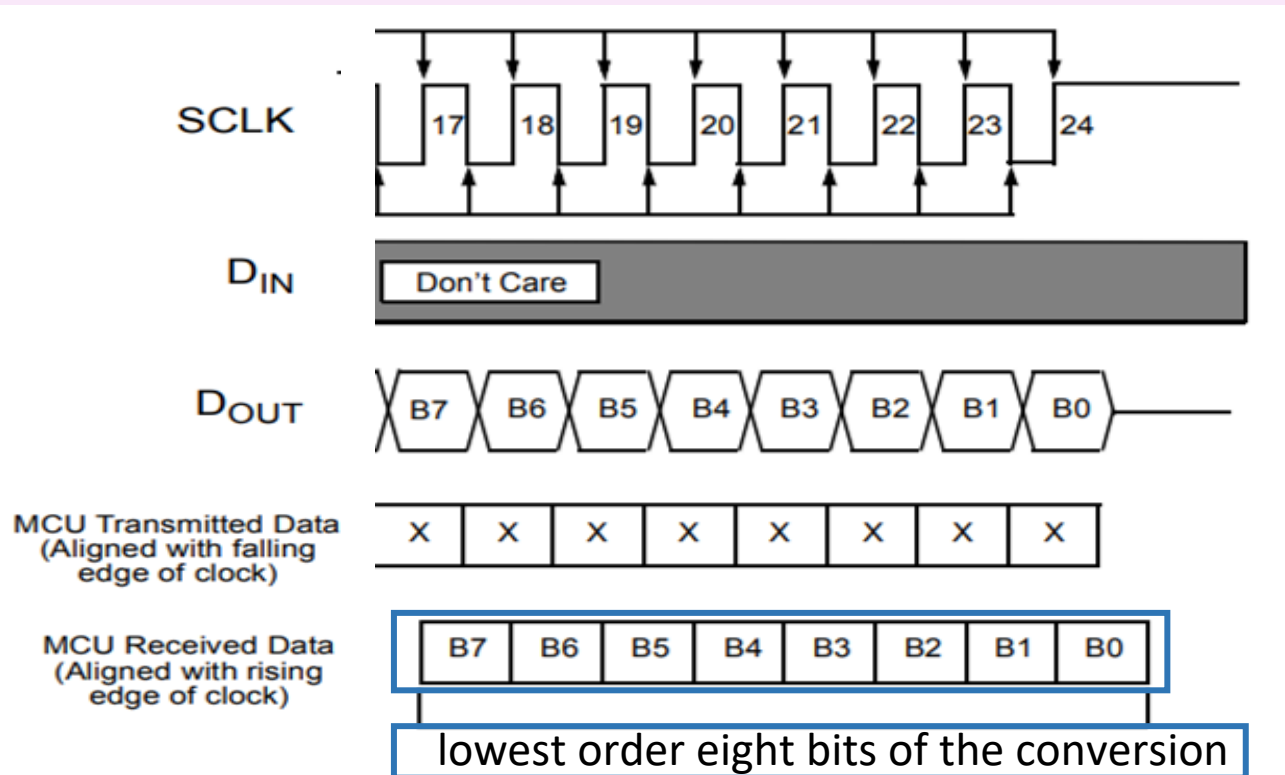
- First byte transmitted to the A/D converter contains seven leading zeros before the start bit.
- This way produces the output 12 bits to fall in positions easily manipulated by the MCU.

8 Bit MCU Transmission Arrangement



- MCU receive buffer will contain three unknown bits until the null bit is clocked out.
- Second byte consists of Null bit and the highest order four bits of the conversion.

8 Bit MCU Transmission Arrangement



- Third byte consists of lowest order eight bits of the conversion.
- Easier manipulation of the converted data can be obtained by using this method.

Reference

- https://global.sharp/products/device/lineup/data/pdf/datasheet/gp2y0a51sk_e.pdf
- [https://www.researchgate.net/publication/279171564 Novel Method of Detecting Movement of the Interference Fringes Using One-Dimensional PSD](https://www.researchgate.net/publication/279171564_Novel_Method_of_Detecting_Movement_of_the_Interference_Fringes_Using_One-Dimensional_PSD)
- <https://ww1.microchip.com/downloads/en/DeviceDoc/21034F.pdf>
- https://en.wikipedia.org/wiki/Analog-to-digital_converter

IR Sensor Operating Interval

■Absolute maximum ratings

(Ta=25°C)

Parameter	Symbol	Ratings	Unit
Supply voltage	Vcc	-0.3 to +7	V
Output terminal voltage	Vo	-0.3 to Vcc+0.3	V
Operating temperature	Topr	-10 to +60	°C
Storage temperature	Tstg	-40 to +70	°C

■Operating supply voltage

Symbol	Rating	Unit
Vcc	4.5 to 5.5	V

IR Sensor Measuring Interval

■Electro-optical Characteristics

(Ta=25°C, Vcc=5V)

Parameter	Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Measuring distance range	ΔL	(Note 1)	2	-	15	cm
Output terminal voltage	V_o	L=15cm (Note 1)	0.25	0.4	0.55	V
Output voltage difference	ΔV_o	Output change at L change (15cm → 2cm) (Note 1)	1.35	1.65	1.95	V
Average supply current	I_{cc}	L=15cm (Note 1)	-	12	22	mA

※L : Distance to reflective object

(Note 1) Using reflective object : White paper

(Made by Kodak Co., Ltd. gray cards R-27 • white face, reflective ratio ; 90%)

Microprocessor Systems

Lab7-2 MLX90614 IR Thermometer

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NTHU EE



Introduction

- MLX90614 is an Infrared thermometer for noncontact temperature measurements
- MLX90614 is built with 2 chips
 - Infrared thermopile detector MLX81101
 - Single conditioning ASSP MLX90302

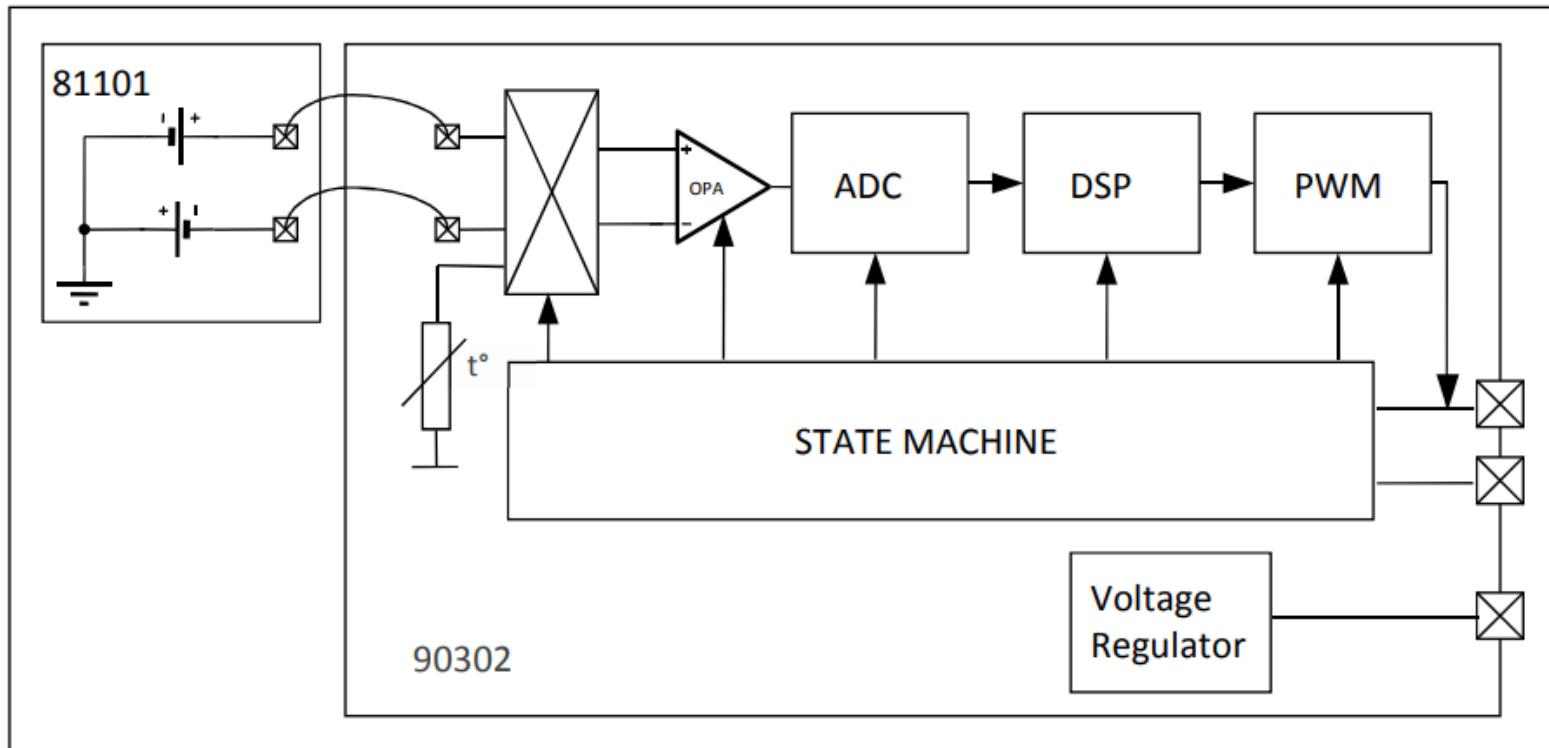
Features

- Measuring ambient temperature (T_a) and object temperature (T_o)
- Factory calibrated in wide temperature range
 - $-40^{\circ}\text{C} \sim +125^{\circ}\text{C}$ for sensor temperature (T_a)
 - $-70^{\circ}\text{C} \sim +380^{\circ}\text{C}$ for object temperature (T_o)
- Measurement resolution of 0.02°C
- Sleep mode for reduced power consumption

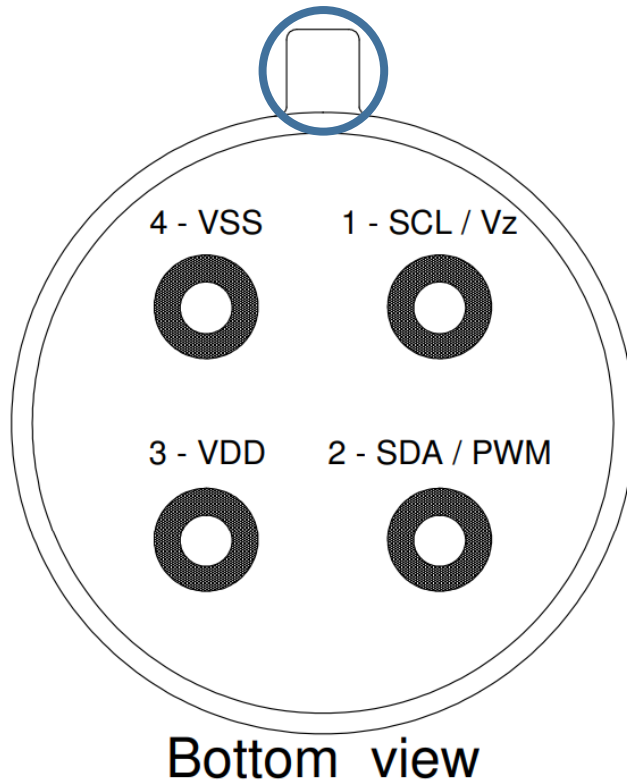
Applications of MLX90614

- Non-contact thermometer (Lab project)
- Temperature Measurement of moving objects
- Industrial Thermal Gun
- Human Body Temperature Measurement
- Home/Office Temperature Control
- Livestock Monitoring
- Movement Detection

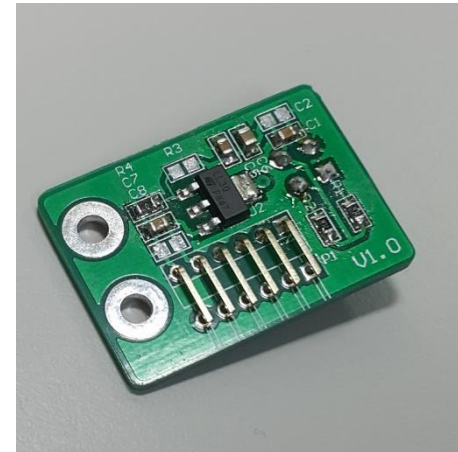
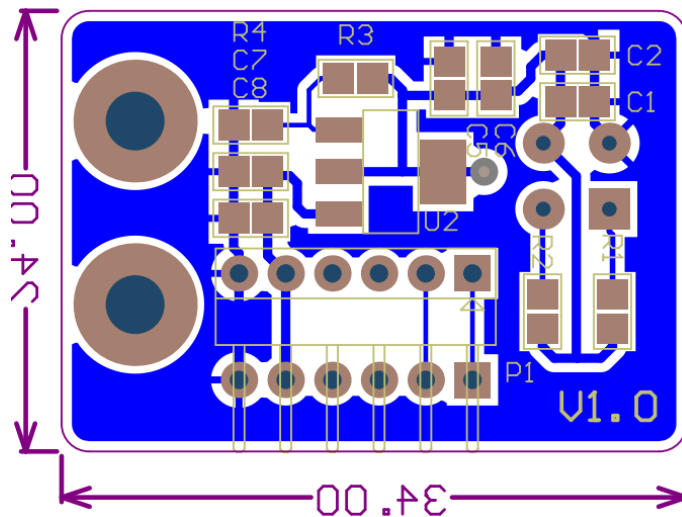
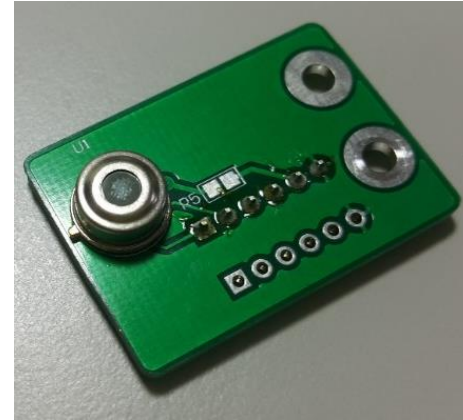
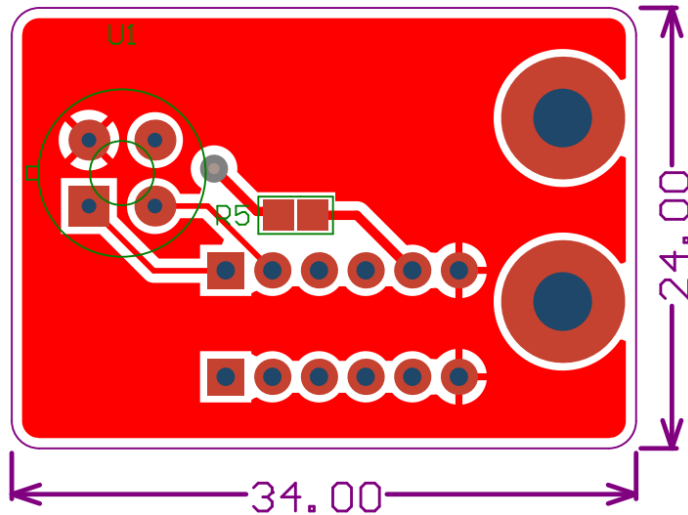
Block Diagram



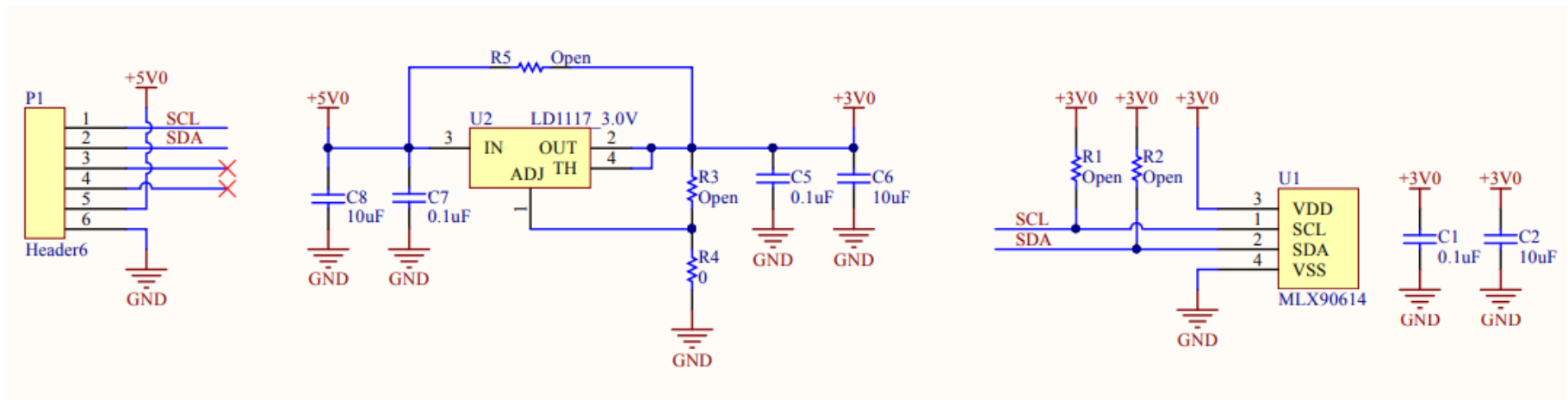
MLX90614



MLX90614 With PCB

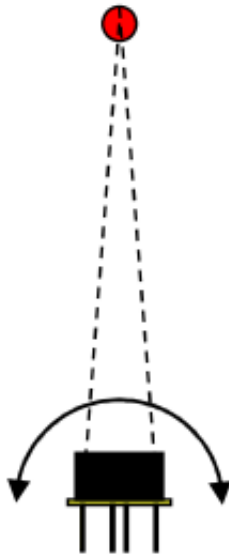


Connection with MCU

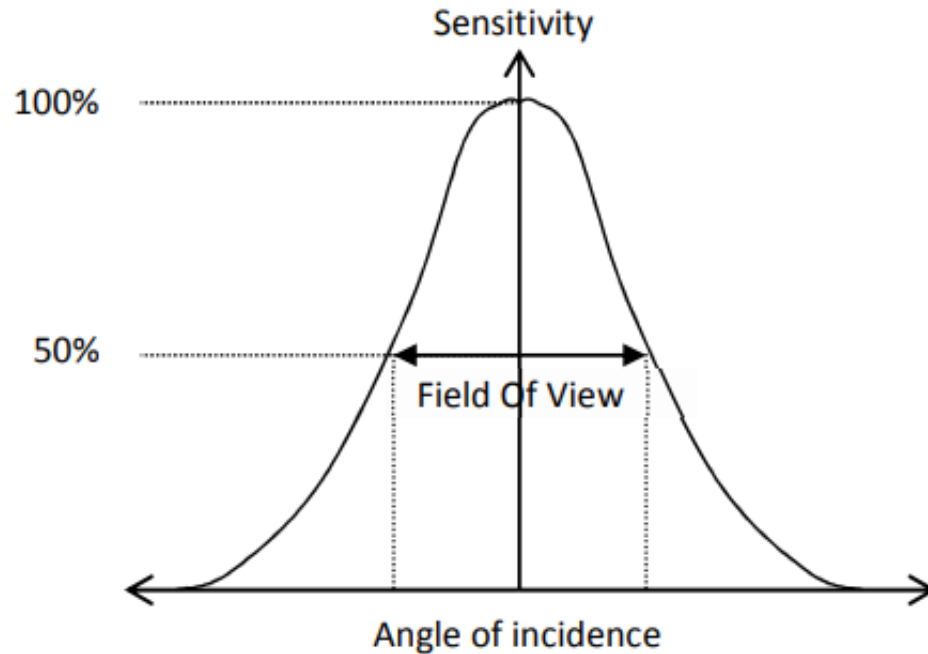


Field of View

Point heat source



Rotated sensor



How Field Of View For Infrared Temperature Sensors Is Determined

<https://youtu.be/vpN-myfTAOs>

SMBus Compatible 2-wire Interface

- Slave address: 0x5A

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
SMBus compatible 2-wire interface ²						
Input high voltage	$V_{IH}(Ta,V)$	Over temperature and supply	VDD-0.1			V
Input low voltage	$V_{IL}(Ta,V)$	Over temperature and supply			0.6	V
Output low voltage	V_{OL}	Over temperature and supply, $I_{sink} = 2mA$			0.25	V
SCL leakage	$I_{SCL,leak}$	$V_{SCL}=3V, Ta=+85^{\circ}C$			20	μA
SDA leakage	$I_{SDA,leak}$	$V_{SDA}=3V, Ta=+85^{\circ}C$			0.25	μA
SCL capacitance	C_{SCL}				10	pF
SDA capacitance	C_{SDA}				10	pF
Slave address	SA	Factory default		5A		hex

Electrical Specifications

- Typical Supply current = 1.3mA
- Sleep mode current = 2.5uA

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
Supplies						
External supply	V_{DD}		2.6	3	3.6	V
Supply current	I_{DD}	No load		1.3	2	mA
Supply current (programming)	I_{DDpr}	No load, erase / write EEPROM operations		1.5	2.5	mA
Sleep mode current	I_{sleep}	no load	1	2.5	5	μ A
Sleep mode current	I_{sleep}	Full temperature range	1	2.5	6	μ A

EEPROM

- Size : 32x16 bits (32 register , 16 bit/register)

EEPROM (32X16)		
Name	Address	Write access
$T_{o_{max}}$	0x00	Yes
$T_{o_{min}}$	0x01	Yes
PWMCTRL	0x02	Yes
Ta range	0x03	Yes
Emissivity	0x04	Yes
Config Register1	0x05	Yes
Melexis reserved	0x06	No
...	...	...
Melexis reserved	0x0D	No
SMBus address (LSByte only)	0x0E	Yes
Melexis reserved	0x0F	Yes
Melexis reserved	0x10	No
...	...	...
Melexis reserved	0x18	No
Melexis reserved	0x19	Yes
Melexis reserved	0x1A	No
Melexis reserved	0x1B	No
ID number	0x1C	No
ID number	0x1D	No
ID number	0x1E	No
ID number	0x1F	No

RAM

- Size : 32x16 bits (32 register , 16 bit/register)

RAM (32x16)		
Name	Address	Read access
Melexis reserved	0x00	Yes
...	...	...
Melexis reserved	0x03	Yes
Raw data IR channel 1	0x04	
Raw data IR channel 2	0x05	
T _A	0x06	Yes
T _{OBJ1}	0x07	Yes
T _{OBJ2}	0x08	Yes
Melexis reserved	0x09	Yes
...	...	...
Melexis reserved	0x1F	Yes

Command

Five bits correspond to 32 registers

Opcode	Command
000x xxxx*	RAM Access
001x xxxx*	EEPROM Access
1111_0000**	Read Flags
1111_1111	Enter SLEEP mode

- *Note *: The xxxxx represent the 5 LSBits of the memory map address to be read / written*
- *Note **: Behaves like read command. The difference is that the read flags does not have a repeated start bit*

Bit Transfer

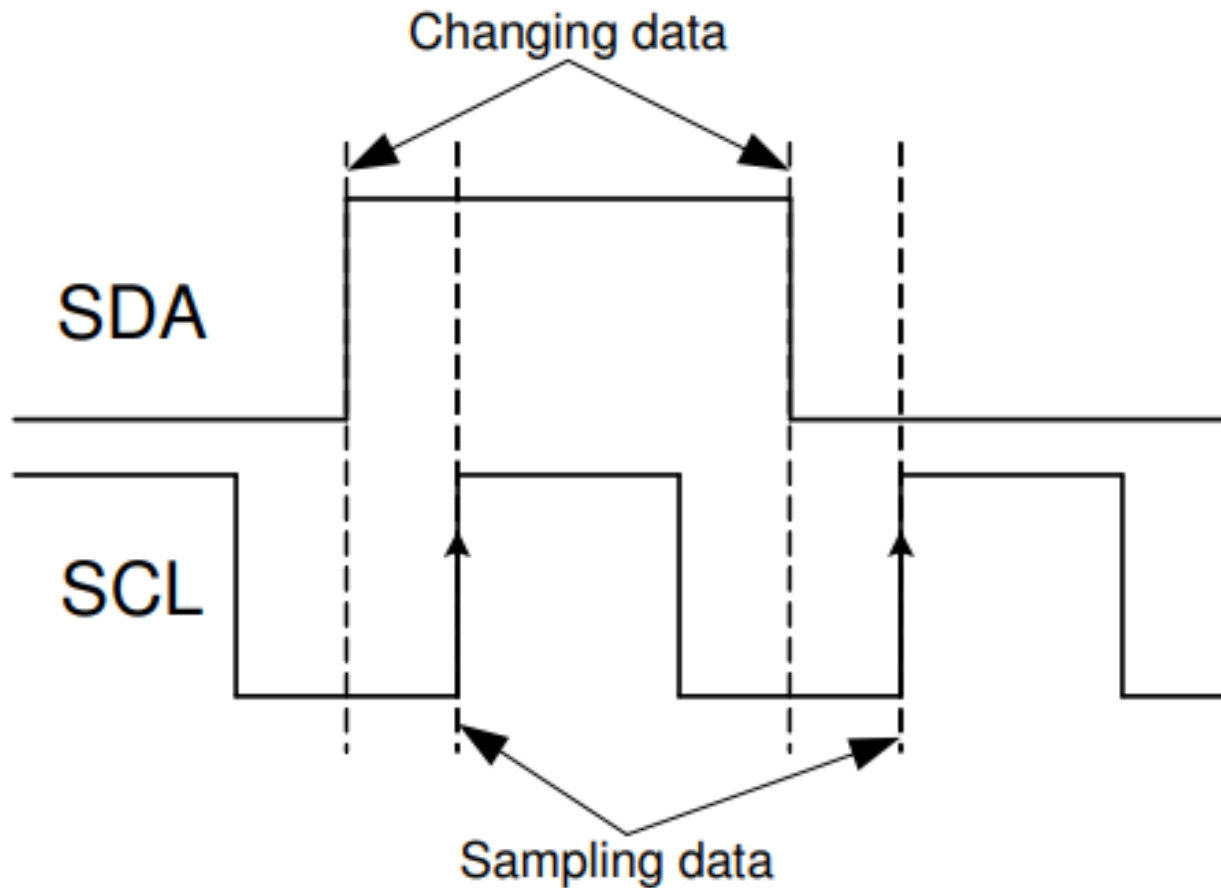
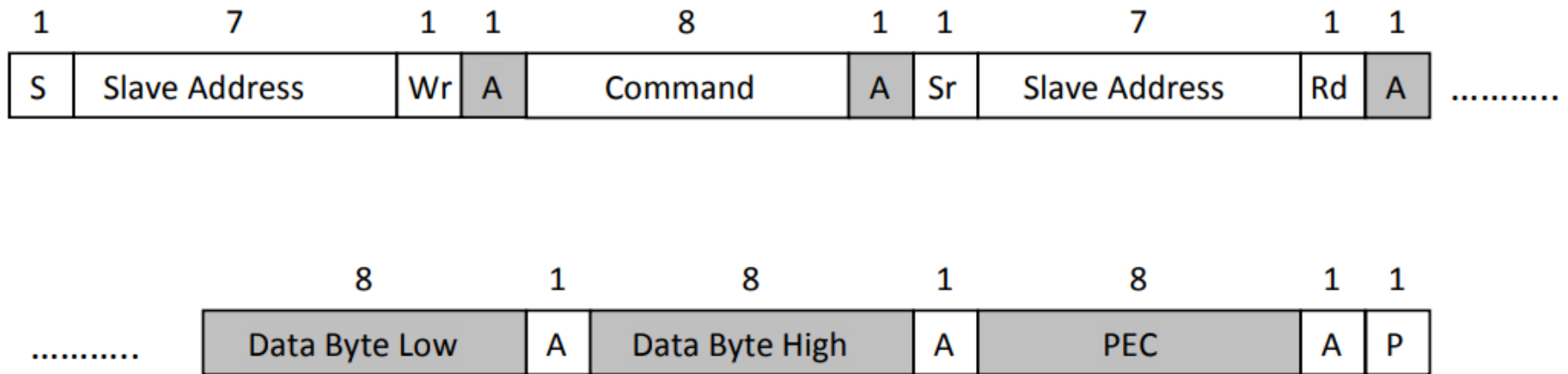


Figure 7: Recommended timing on SMBus

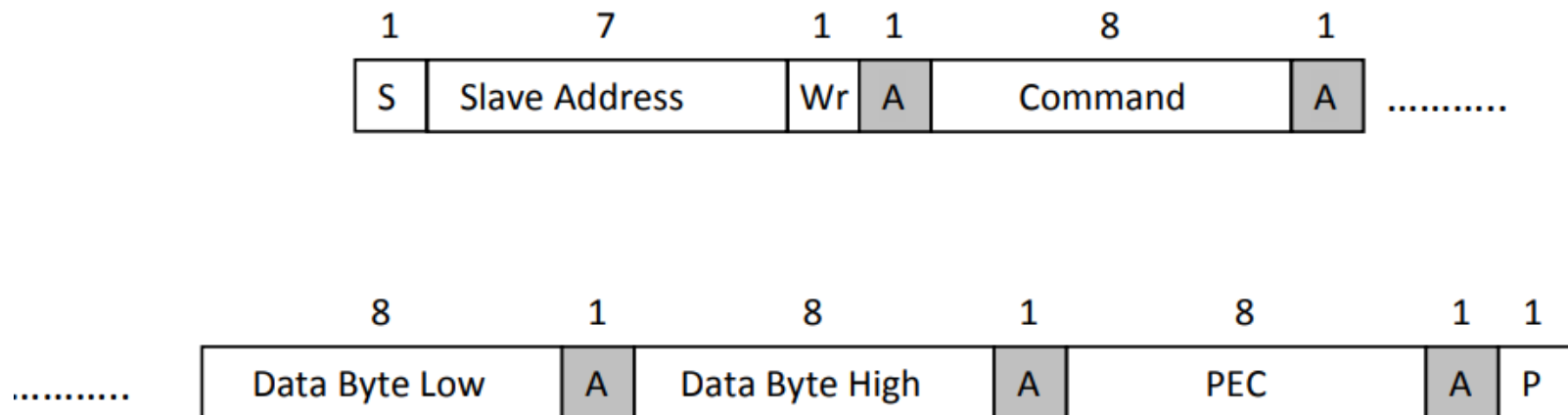
Read Word

- Access RAM, read T_a , T_o



Write Word

- Write data in EEPROM



SMBus Communication Example

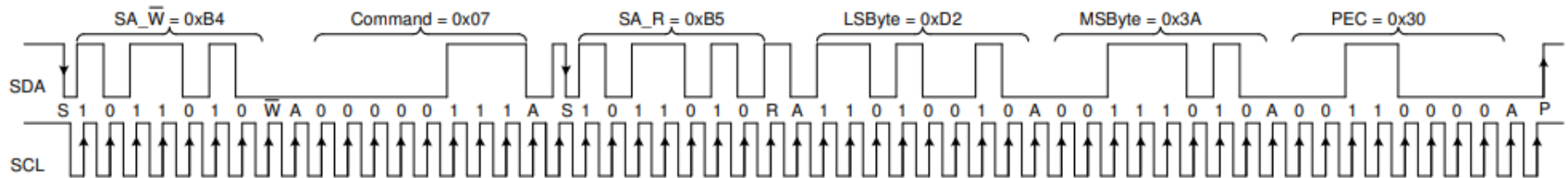


Figure 8: Read word format (SA=0x5A, read RAM=0x07, result=0x3AD2, PEC=0x30)

SMBus Communication Example

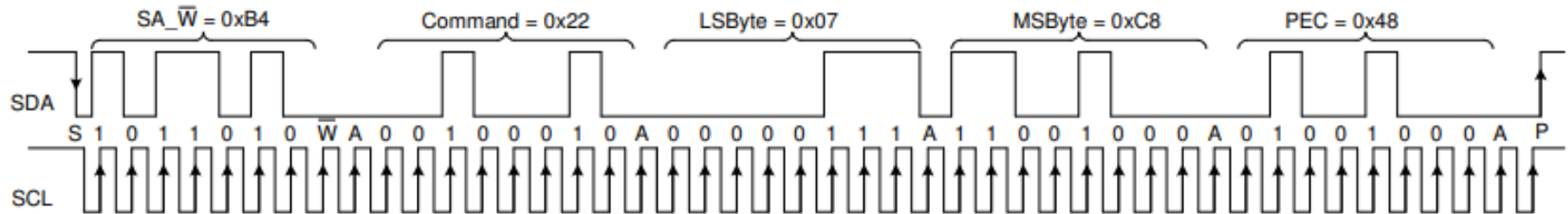


Figure 9: Write word format (SA=0x5A, write EEPROM=0x02, data=0xC807, PEC=0x48)

Command 0x22 = 0010_0010

Computation of Ambient and Object Temperatures

- Thermopile output signal: $V_{IR} (T_a, T_o) = A \times (T_o^4 \times T_a^4)$
 - T_o : Absolute object temperature
 - T_a : Sensor die absolute temperature
 - A : Sensitivity
- T_a, T_o are derived from RAM
 - $T_a [^{\circ}\text{C}] = T_{areg} \times 0.02 - 273.15$
 - $0x06 = 0x2DE4$ (11748d) corresponds to -38.2°C
 - $0x06 = 0x4DC4$ (19908d) corresponds to $+125^{\circ}\text{C}$
 - $T_o [^{\circ}\text{C}] = T_{oreg} \times 0.02 - 273.15$
 - $0x27AD$ corresponds to -70°C
 - $0x7FFF$ corresponds to 382.19°C

Reference

- <https://www.melexis.com/zh/documents/documentation/datasheets/datasheet-mlx90614>
- <https://www.mouser.tw/datasheet/2/268/mic5270-1891236.pdf>